

Sec 3.10 – Theorems about Differentiable Functions

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

We often want to infer a global conclusion, like the function f is increasing on an interval from local information, like f' is positive at each point on an interval.

The following theorem relates the average rate of change of a function on an interval (global information) to the instantaneous rate of change at a point (local information).

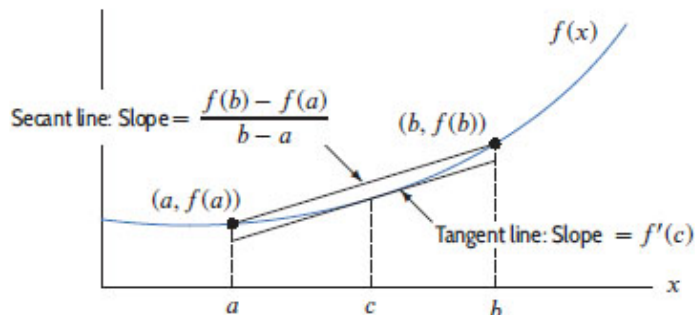
The Mean Value Theorem

Theorem (MVT)

If f is continuous on $a \leq x \leq b$ and differentiable on $a < x < b$, then there exists a number c , with $a < c < b$, such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

In other words, $f(b) - f(a) = f'(c)(b - a)$.



The Mean Value Theorem (MVT) tells us that the point $x=c$ exists, but does not tell us how to find c .

Suppose $f(x) = x^3 - 2x^2 + 3x + 5$

- Show $f(x)$ satisfies the hypotheses of the MVT for the interval $[1, 2]$.
- What is the conclusion of the MVT for the interval $[1, 2]$?

Solution:

$f(x) = x^3 - 2x^2 + 3x + 5$ is both continuous and differentiable on all the real numbers since it is a polynomial and hence satisfies the hypotheses.

The conclusion from the MVT is that there is some c such that $1 < c < 2$ and $f'(c) = \frac{f(2) - f(1)}{2 - 1} = \frac{11 - 7}{2 - 1} = 4$.

This doesn't tell us where c is - but we could solve and find c :

$f'(x) = 3x^2 - 4x + 3 = 4$ implies that $3x^2 - 4x - 1 = 0$ and we can use the quadratic formula to find that $x = \frac{2 \pm \sqrt{7}}{3}$, where $x = \frac{2 + \sqrt{7}}{3} \approx 1.55$, which is the desired c .

Given $ax^2 + bx + c = 0$, the quadratic formula is: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, and applying it here gives:

$$x = \frac{4 \pm \sqrt{4^2 - 4 \cdot 3 \cdot (-1)}}{2 \cdot 3} = \frac{4 \pm \sqrt{4^2 + 12}}{6} = \frac{4 \pm \sqrt{28}}{6} = \frac{2 \pm \sqrt{7}}{3} \text{ (note we drop the negative in the } \pm \text{ since the interval is } [1, 2]), \text{ so}$$

$$x \approx 1.55$$

Partner up

Suppose you know $g(x)$ is differentiable, $g(2) = -4$ and $g'(x) \geq 3$ for $2 \leq x \leq 5$. What can you infer about $g(5)$?

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Since g is differentiable, it is also continuous and MVT applies. That is, using $a = 2$ and $b = 5$, $g(5) - g(2) = g'(c)(5 - 2)$. Since $g'(c) \geq 3$ no matter where c is chosen, we know that $g(5) - g(2) \geq 3(5 - 2) = 9$ or $g(5) \geq 9 + g(2) = 9 + (-4) = 5$. Therefore, $g(5) \geq 5$.

Next, we present the **Racetrack Principle**:

Theorem (3.10)

Suppose that g and h are continuous on $a \leq x \leq b$ and differentiable on $a < x < b$, and that $g'(x) \leq h'(x)$ for $a < x < b$.

- If $g(a) = h(a)$, then $g(x) \leq h(x)$ for $a \leq x \leq b$.
- If $g(b) = h(b)$, then $g(x) \geq h(x)$ for $a \leq x \leq b$.

We can interpret this as: If we think of $g(x)$ and $h(x)$ as the positions of two racehorses at time x , with horse h always moving faster than horse g .

If they start together, horse h is ahead during the whole race. If they finish together, horse g was ahead during the whole race.

Explain why $e^x \geq 1 + x$ for $x \geq 0$.

Graphically:

e^x is concave up everywhere and the equation of the tangent line at the point $(0,1)$ is $y = x + 1$, and e^x always is above its tangent line.

We can also prove this using the Racetrack Principle:

Let $g(x) = 1 + x$ and $h(x) = e^x$

$g(0) = h(0) = 1$, and we know $g'(x) = 1$ and $h'(x) = e^x$, which means $g'(x) \leq h'(x)$ for $x \geq 0$.

So by the Racetrack Principle, for $a = 0$ we have $g(x) \leq h(x)$, or $1 + x \leq e^x$.

